SHED MEASUREMENT WORKING PAPER

Emergency Buffers, Borrowed Resilience, and Household Hardship

Evidence from SHED 2015–2025

Arne Kotte

AUTHOR
Arne Kotte

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Arne Kotte

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Abstract

This paper consolidates a set of SHED emergency-buffer analyses into one measurement paper on adult respondents' household financial resilience. The central claim is deliberately narrow: income, emergency savings, and generic credit-use indicators are incomplete measures because they collapse distinct adult-respondent states. A respondent with liquid resources, a respondent using paid-in-full bridge credit, a respondent revolving a card balance, a respondent using other emergency borrowing, and a respondent unable to pay a $400 emergency are not in the same financial position. Using Federal Reserve Survey of Household Economics and Decisionmaking (SHED) public-use microdata and harmonized repeated cross-sections from 2015–2025, the paper documents three layers of evidence. First, published and replicated SHED indicators show that emergency-payment capacity improved through 2021 and then plateaued or reversed by 2025. Second, 2025 respondent-level profiles show large hardship gradients: strict hardship is 24.7 percent among paid-in-full bridge-credit respondents, 31.4 percent among cash/checking/savings respondents, 67.9 percent among revolving-credit respondents, 79.3 percent among other-borrowing respondents, 85.1 percent among respondents unable to pay, and 78.8 percent in the pooled liquidity-exclusion/unclassified group. Third, non-exclusive response models, first-pass clustering, benchmark screens, and multi-year validation support the broad ordering while preserving an explicit public-use inference boundary. The paper is descriptive, not causal. Its contribution is a transparent measurement scaffold for distinguishing liquidity sufficiency, liquidity bridges, debt-financed distress, and liquidity exclusion in household-resilience analysis.

1. Introduction

Household financial resilience is often reduced to income, emergency savings, or debt. Those measures are useful, but they do not tell the whole story. Income is a flow. Emergency savings are a liquid stock. Credit can be a convenience instrument, a short-term bridge, distress borrowing, or unavailable when needed. Hardship outcomes such as bill pressure, external help, and medical debt describe realized strain.

This paper consolidates three related SHED research artifacts into one empirical measurement argument. The first artifact used published SHED indicators to show the emergency-buffer time pattern. The second built a public-use microdata replication and harmonization scaffold. The third developed a 2025 emergency-financing profile taxonomy. The flagship contribution is to combine these layers: public indicators motivate the problem, replication/harmonization establishes the data path, and the 2025 profile decomposition supplies the main measurement contribution.

The paper asks a practical question: when adults face a $400 emergency expense, what financial state does their reported payment path reveal? The answer is not simply “has credit” or “does not have credit.” Paid-in-full credit use looks much closer to liquidity sufficiency than to revolving debt or emergency borrowing. Low-buffer non-use may signal exclusion rather than resilience. The measurement problem is therefore to separate liquid sufficiency, liquidity bridges, debt-financed distress, and liquidity exclusion without claiming that any one state causally produces or prevents hardship.

The estimates below are descriptive adult-respondent associations. The paper does not estimate the causal effect of credit or savings on hardship. It does not present a validated prediction index. It instead provides an institutional-grade measurement scaffold: a transparent way to classify emergency-financing states and test whether the resulting ordering is stable across outcomes, coding choices, simpler screens, and repeated SHED cross-sections.

2. Conceptual frame

The guiding concept is household shock absorption. A financially resilient adult or household can absorb an emergency without converting the shock into arrears, medical debt, outside help, high-cost borrowing, or inability to pay. In practice, SHED observes reported adult responses rather than full balance sheets. The analysis therefore separates three objects:

1. **Liquid-buffer exposure:** whether the respondent reports immediate $400 cash/equivalent capacity and three-month emergency savings.
2. **Emergency-financing behavior:** whether the respondent would use cash/checking/savings, paid-in-full card credit, revolving credit, other borrowing, or would be unable to pay.
3. **Hardship outcomes:** bill pressure, external help with expenses, medical debt, financial vulnerability, and related markers.

Four states organize the interpretation:

1. **Liquidity sufficient:** the respondent can use cash, checking, savings, or similar liquid resources.
2. **Liquidity bridge:** the respondent uses credit as short-term payment smoothing and can pay it in full.
3. **Debt-financed distress:** the respondent would revolve credit or use other emergency borrowing.
4. **Liquidity exclusion:** the respondent cannot pay, lacks a listed financing path, or appears constrained despite low buffers.

These are measurement states, not welfare rankings or treatment effects. Their value is interpretive: they prevent paid-in-full credit, revolving balances, other borrowing, and inability to pay from being collapsed into one generic credit-use variable.

3. Data, weights, and public-use inference boundary

The analysis uses Federal Reserve SHED public-use files and public SHED published indicators. The unit is an adult respondent. Some variables describe household expenses or household circumstances, but weighted counts should be read as weighted adults, not directly as sampled households.

The 2025 pooled profile table uses final respondent weights and 12,934 adults with positive weights and observed profile/outcome fields, representing about 264.5 million weighted adults. Adjusted 2025 models use 12,932 adults after control-variable completeness restrictions. The repeated-cross-section validation uses harmonized SHED variables from 2015–2025 where comparable items are available.

A key limitation is survey-design inference. The public-use files inspected for this project include final cross-sectional weights and population weights but not public replicate weights, strata, or PSU variables. Reported confidence intervals and model standard errors therefore use final-weight/Kish and robust-WLS approximations. They are useful for scale, screening, and internal comparison, but they are not final design-consistent survey inference.

4. Published-indicator and harmonized background

Published SHED indicators motivate the microdata analysis. Emergency-payment capacity improved through 2021 and then plateaued or reversed. The pattern matters because a single cross-section could otherwise be mistaken for a timeless fact rather than the latest point in a repeated public measurement series.

Table 1. Emergency-buffer milestones, all adults. Values are percent of adults. The $400 measure is ability to cover the expense completely using cash or its equivalent. The 3-month measure is having emergency savings to cover three months of expenses.

Year	$400 cash/equivalent	Cannot cover $400 with cash/equiv.	Has 3-month emergency savings	Lacks 3-month emergency savings
2013	50	50	—	—
2015	54	46	47	53
2019	63	37	53	47
2021	68	32	59	41
2022	63	37	54	46
2024	63	37	55	45
2025	63	37	55	45

Table 1 shows the public-indicator background. The $400 cash/equivalent measure rose from 63 percent in 2019 to 68 percent in 2021 and then returned to 63 percent by 2025. Three-month emergency savings rose from 53 percent in 2019 to 59 percent in 2021 and then fell to 55 percent by 2025. The broader lesson is that post-2021 emergency-buffer conditions did not continue improving.

Table 2. Annual weighted rates with official benchmark. Official cash/equiv $400 is the Federal Reserve published SHED report series. Constructed cash/equiv is derived from microdata component responses and is shown only as a diagnostic, not as the same object.

Year	Official cash/equiv $400	Constructed/direct diagnostic	Card over time	Unable $400 now	Three-month savings	Doing okay	Financially vulnerable
2015	54.0	66.0	17.4	13.9	48.1	68.6	31.4
2016	56.0	68.7	20.0	12.1	48.7	70.0	30.0
2017	59.0	71.6	17.9	12.0	50.5	73.7	26.3
2018	61.0	69.1	15.6	12.4	51.6	74.9	25.1
2019	63.0	72.0	15.8	12.6	53.3	75.4	24.6
2020	64.0	64.6	15.6	12.6	55.4	75.1	24.9
2021	68.0	67.7	13.9	10.8	59.1	77.7	22.3
2022	63.0	65.0	15.8	13.1	54.1	73.1	26.9
2023	63.0	63.3	15.6	12.8	53.8	72.2	27.8
2024	63.0	62.7	15.2	13.0	55.0	72.9	27.1
2025	63.0	63.1	15.3	12.2	54.8	72.9	27.1

Table 2 links the published-indicator lane to the harmonized microdata lane. The official $400 series, the constructed/direct diagnostic, card-over-time responses, inability-to-pay responses, three-month savings, and financial-vulnerability measures provide the background for the 2025 profile analysis. The official and constructed/direct diagnostics should not be treated as identical before 2020, but the post-2020 alignment gives confidence that the microdata pipeline is tracking the relevant concepts.

5. Construct definitions and overlap control

The core measurement risk is mechanical overlap. If the hardship outcome includes inability to pay $400 or credit constraint, then emergency-financing profiles can mechanically generate hardship. The main outcome therefore uses a strict non-overlap definition: bill pressure, external help with expenses, or medical debt. Broader composites remain useful, but the strict outcome is the main anchor.

Table 3. SHED hardship-measurement construct map. Revised construct definitions. The primary outcome is a strict non-overlap hardship measure that excludes the emergency-capacity item, credit constraint, and the broad financial-wellbeing summary.

Construct	Variable	Definition	Role	Caveat
Low liquid buffer	low_buffer_both	Cannot cover $400 with cash/equivalent and lacks three months emergency savings	Exposure	Not a bank-balance measure; SHED response-based
Credit reliance	credit_reliance_any	Would pay $400 by credit over time or another emergency borrowing response	Moderator/exposure	Broad first-pass emergency-financing screen; excludes credit-constraint outcome to reduce mechanical overlap
Emergency-financing profile	emergency_financing_profile	Mutually exclusive profile: unable to pay, revolving credit, other borrowing, constrained non-use, bridge credit paid in full, cash/checking/savings, or other	Decomposition	Priority-coded descriptive taxonomy; not a welfare ranking
Primary strict hardship	hardship_strict_nonoverlap	Bill pressure, external help, or medical debt	Primary outcome	Excludes credit constraint, financial-wellbeing summary, and unable-pay-$400
Broad hardship composite	hardship_composite	Financial vulnerability, bill pressure, external help, credit constraint, or medical debt	Robustness outcome	Excludes unable-pay-$400 but still overlaps with constrained-nonuse profile through credit constraint
Bill pressure	bill_pressure	Did not pay all non-card bills in full or had difficult bills	Outcome	2025 build; harmonization needed for multi-year work
External help	fs21_any_external_help	Received outside-household help with listed expenses	Outcome	Descriptive support reliance
Medical debt	medical_debt	Currently has debt from medical care	Outcome	SHED self-report; timing and mechanism may differ from contemporaneous bill pressure

The low-buffer exposure is strict: the respondent both cannot cover a $400 emergency with cash/equivalent and lacks three months of emergency savings. Emergency credit reliance is a broad screen for revolving credit or other borrowing. The emergency-financing profile is the main decomposition.

6. Main 2025 emergency-financing profile decomposition

The flagship empirical table separates cash/checking/savings, paid-in-full bridge credit, revolving credit, other emergency borrowing, inability to pay, and pooled liquidity-exclusion/unclassified states.

Table 4. Pooled emergency-financing profiles and hardship, SHED 2025. Small diagnostic cells are pooled into `Liquidity exclusion / unclassified` for main-text interpretation. Counts sum to 12,934 adult respondents with observed profile/outcome fields; adjusted models below use 12,932 adults after control-variable completeness restrictions. Profiles are descriptive groups, not causal treatments or welfare rankings.

Emergency-financing profile	N	Weighted pop.	Strict hardship % [95% approx CI]	Broad composite %	Low buffer %	Financially vulnerable %	Bill pressure %	Credit constrained %	Medical debt %
Cash/checking/savings	3,754	76.2M	31.4 [29.8, 33.0]	39.7	0.5	12.0	12.9	11.7	13.0
Bridge credit paid in full	4,704	92.1M	24.7 [23.4, 26.0]	30.3	0.3	7.8	8.7	6.2	8.1
Revolving credit for $400	1,833	38.3M	67.9 [65.6, 70.2]	80.3	74.4	47.6	45.9	30.8	32.4
Other emergency borrowing	943	21.5M	79.3 [76.5, 82.1]	87.3	82.7	52.8	56.0	36.1	28.2
Unable to pay $400	1,530	32.3M	85.1 [83.2, 87.0]	93.5	92.7	73.4	72.9	42.7	35.3
Liquidity exclusion / unclassified	170	4.1M	78.8 [72.2, 85.4]	86.8	90.0	48.2	62.8	32.3	27.1

Table 4 is the core result. Strict hardship is 24.7 percent among bridge-credit-paid-in-full respondents and 31.4 percent among cash/checking/savings respondents. It rises to 67.9 percent among revolving-credit respondents, 79.3 percent among other emergency borrowing respondents, 85.1 percent among respondents unable to pay $400, and 78.8 percent in the pooled liquidity-exclusion/unclassified group. This is the central measurement contribution: “credit use” is not one state.

The bridge-credit finding should not be read causally. Paid-in-full card users may have better credit access, more stable income, higher financial sophistication, or lower unobserved risk. The correct conclusion is narrower and stronger: paid-in-full credit identifies a different financial state from revolving credit and distress borrowing.

7. Multiple-response and non-exclusive validation

SHED respondents can select multiple ways they would handle a $400 emergency. Any mutually exclusive taxonomy therefore requires a priority rule. The flagship paper treats that as a feature to validate, not a nuisance to hide.

Table 5. Non-exclusive emergency-response dummy model for strict hardship, SHED 2025. Weighted LPM coefficients in percentage points with approximate robust standard errors in parentheses. Emergency-response indicators are not mutually exclusive. Controls include income, education, age, race/ethnicity, housing tenure, employment, children, and region. N=12,932; weighted adult population=264.5M; R^2=0.295. Descriptive validation benchmark, not causal inference.

Non-exclusive response / exposure	Coefficient pp (SE)	CI low	CI high
Paid-in-full card response	-7.6 (1.2)	-9.9	-5.3
Revolving card response	11.9 (1.4)	9.2	14.7
Cash/checking/savings response	-5.5 (1.1)	-7.7	-3.4
Other emergency borrowing response	13.3 (1.4)	10.6	16.0
Unable-to-pay response	14.2 (1.5)	11.2	17.1
Low buffer	14.2 (1.6)	11.1	17.4

Table 5 addresses the priority-rule artifact concern. Paid-in-full card and cash/checking/savings responses are negatively associated with strict hardship after controls, while revolving card, other borrowing, inability-to-pay response, and low-buffer status are positively associated with strict hardship. Because these indicators are non-exclusive, the result supports the taxonomy without relying entirely on one priority-coded profile assignment.

8. Data-driven taxonomy validation

The first-pass clustering validation uses emergency-response, low-buffer, and credit-constraint indicators only; hardship outcomes are held out until after cluster assignment. This is not a final latent-class model, but it is a useful test of whether response patterns themselves recover the conceptual states.

Table 6. Data-driven emergency-financing clusters, SHED 2025. Weighted k-means clusters use emergency-response, low-buffer, and credit-constraint indicators only; hardship outcomes are held out and reported after clustering. Cluster labels are interpretive summaries of centroid rates, not new causal classes.

Cluster	Interpretive label	N	Weighted pop.	Strict hardship % [95% approx CI]	Financially vulnerable %	Bill pressure %	Cash response %	Paid-full card %	Revolving card %	Other borrowing %	Unable-to-pay %	Low buffer %	Credit constrained %
C1	Empirical bridge-credit cluster	3,719	72.9M	25.7 [24.2, 27.2]	8.8	10.2	0.0	99.6	1.1	0.9	0.1	0.3	6.6
C2	Empirical cash-plus-bridge cluster	1,269	25.2M	29.5 [26.8, 32.2]	8.7	8.9	100.0	100.0	10.7	6.3	0.5	7.5	7.2
C3	Empirical cash/liquid-sufficiency cluster	3,910	79.5M	32.9 [31.3, 34.5]	12.8	13.8	99.8	0.0	2.8	1.3	0.3	0.5	12.2
C4	Empirical revolving / borrowing cluster	2,576	56.0M	74.8 [72.9, 76.6]	53.7	54.3	15.7	6.4	60.3	48.7	1.8	87.6	35.5
C5	Empirical unable-to-pay / exclusion cluster	1,458	30.9M	84.9 [82.9, 86.9]	73.6	72.8	2.6	0.4	3.4	16.0	100.0	93.8	42.5

Table 6 supports the broad taxonomy. The empirical bridge-credit cluster has strict hardship of 25.7 percent, the cash-plus-bridge cluster 29.5 percent, the cash/liquid-sufficiency cluster 32.9 percent, the revolving/borrowing cluster 74.8 percent, and the unable-to-pay/exclusion cluster 84.9 percent. Since hardship outcomes are held out of clustering, the ordering is not simply outcome sorting.

Table 7. Crosswalk from priority-coded profiles to data-driven clusters, SHED 2025. Cells show the weighted percent of each pooled priority-coded profile assigned to each empirical cluster. This is a robustness diagnostic: high concentration supports the author-imposed taxonomy, while dispersion identifies mixed-response profiles.

Priority-coded pooled profile	C1	C2	C3	C4	C5
Cash/checking/savings	0.0	0.0	100.0	0.0	0.0
Bridge credit paid in full	77.2	22.8	0.0	0.0	0.0
Revolving credit for $400	2.0	6.9	5.8	85.3	0.0
Other emergency borrowing	3.1	6.6	3.2	87.2	0.0
Unable to pay $400	0.1	0.4	0.8	3.2	95.6
Liquidity exclusion / unclassified	7.7	1.5	3.4	87.3	0.0

Table 7 crosswalks author-coded profiles to empirical clusters. Cash/checking/savings maps cleanly to the empirical cash cluster. Bridge credit mostly maps to bridge and cash-plus-bridge clusters. Revolving credit, other emergency borrowing, and inability to pay concentrate in expected distress/exclusion clusters. The pooled liquidity-exclusion/unclassified category is more dispersed, which is why the paper treats it as diagnostic rather than a precise headline class.

9. Benchmark screens

The profile taxonomy should not be sold as a large predictive breakthrough. Its contribution is measurement clarity. The benchmark-screen comparison keeps that humility explicit.

Table 8. Benchmark screens for strict hardship, SHED 2025. Weighted descriptive comparison of simpler screens against the pooled emergency-financing profile taxonomy. Positive/negative rates are unadjusted strict-hardship rates. Model R^2 is from controls plus the listed screen or profile indicators and is included only as secondary descriptive validation.

Screen / profile set	Positive N / model N	Positive pop. / model pop.	Strict hardship if positive %	Strict hardship if negative %	Unadjusted gap pp	Controls + screen/profile R^2
Income below $50k	4,342	88.7M	69.7	33.4	36.3	0.219
Cannot cover $400 with cash/equivalent	4,534	97.5M	76.4	27.5	49.0	0.294
Lacks three months emergency savings	5,606	119.6M	68.1	26.9	41.1	0.266
Low buffer: both $400 and 3-month savings	3,736	80.5M	79.1	30.8	48.3	0.280
Broad emergency credit reliance	3,115	67.0M	74.3	35.8	38.5	0.260
Pooled emergency-financing profile	12,932	264.5M	—	—	—	0.296

Table 8 shows that simpler screens also identify hardship. Income below $50,000, inability to cover $400 with cash/equivalent, lacking three months of savings, the combined low-buffer measure, and broad emergency credit reliance all separate higher-hardship from lower-hardship adults. The pooled emergency-financing profile has an in-sample R^2 close to the $400 cash/equivalent screen. The value added is not a dramatic fit improvement; it is the ability to distinguish bridge credit from distress borrowing and unavailable liquidity.

10. Multi-year validation

The multi-year evidence validates the broad simplified ordering, not every fine 2025 category or the full strict 2025 composite. That limitation is important. External help and medical debt are not harmonized across the full 2015–2025 window, so the long-window validation uses financial vulnerability; bill pressure is available for 2023–2025.

Table 9. Multi-year stability: simplified emergency-financing profiles and financial vulnerability, SHED 2015–2025. Harmonized repeated cross-sections. This longer-window validation uses the financial-vulnerability outcome because strict bill-pressure coverage is available only in 2023–2025.

Year	Profile	N	Financial vulnerability % [95% approx CI]
2015	Cash/checking/savings	1,759	16.3 [14.0, 18.5]
2015	Bridge credit paid in full	1,522	10.7 [8.7, 12.7]
2015	Revolving credit for $400	948	44.5 [40.3, 48.7]
2015	Unable to pay $400	852	72.7 [68.6, 76.7]
2015	Low buffer / no listed financing	472	57.2 [51.1, 63.2]
2015	Other or unclassified	89	29.6 [17.0, 42.2]
2016	Cash/checking/savings	1,551	18.5 [15.9, 21.2]
2016	Bridge credit paid in full	2,396	9.4 [7.8, 10.9]
2016	Revolving credit for $400	1,265	44.9 [41.3, 48.6]
2016	Unable to pay $400	876	75.6 [71.7, 79.6]
2016	Low buffer / no listed financing	453	51.5 [45.1, 57.8]
2016	Other or unclassified	69	27.8 [13.4, 42.3]
2017	Cash/checking/savings	3,669	13.8 [12.3, 15.3]
2017	Bridge credit paid in full	4,245	8.2 [7.1, 9.4]
2017	Revolving credit for $400	2,068	42.8 [40.0, 45.6]
2017	Unable to pay $400	1,408	69.6 [66.4, 72.8]
2017	Low buffer / no listed financing	661	52.1 [47.1, 57.2]
2017	Other or unclassified	136	22.6 [13.4, 31.7]
2018	Cash/checking/savings	3,679	12.3 [11.0, 13.5]
2018	Bridge credit paid in full	3,589	6.8 [5.8, 7.7]
2018	Revolving credit for $400	1,707	43.6 [40.6, 46.5]
2018	Unable to pay $400	1,434	68.3 [65.2, 71.3]
2018	Low buffer / no listed financing	744	48.0 [43.5, 52.6]
2018	Other or unclassified	163	29.9 [20.9, 38.8]
2019	Cash/checking/savings	4,158	11.9 [10.7, 13.0]
2019	Bridge credit paid in full	4,222	6.6 [5.8, 7.5]
2019	Revolving credit for $400	1,758	41.5 [38.8, 44.2]
2019	Unable to pay $400	1,360	69.8 [66.9, 72.7]
2019	Low buffer / no listed financing	611	56.1 [51.3, 61.0]
2019	Other or unclassified	64	23.8 [10.7, 36.8]
2020	Cash/checking/savings	3,945	11.9 [10.8, 13.0]
2020	Bridge credit paid in full	4,249	6.9 [6.1, 7.7]
2020	Revolving credit for $400	1,627	43.1 [40.5, 45.6]
2020	Unable to pay $400	1,245	71.9 [69.2, 74.6]
2020	Low buffer / no listed financing	524	56.9 [52.2, 61.6]
2020	Other or unclassified	58	23.2 [11.3, 35.2]
2021	Cash/checking/savings	4,206	12.3 [11.2, 13.4]
2021	Bridge credit paid in full	4,531	6.0 [5.3, 6.8]
2021	Revolving credit for $400	1,534	40.8 [38.1, 43.6]
2021	Unable to pay $400	1,089	70.5 [67.4, 73.6]
2021	Low buffer / no listed financing	443	53.1 [47.7, 58.4]
2021	Other or unclassified	71	29.2 [16.9, 41.5]

Year	Profile	N	Financial vulnerability % [95% approx CI]
2022	Cash/checking/savings	3,737	12.9 [11.7, 14.1]
2022	Bridge credit paid in full	4,282	8.7 [7.7, 9.6]
2022	Revolving credit for $400	1,705	45.6 [43.0, 48.3]
2022	Unable to pay $400	1,367	70.9 [68.2, 73.6]
2022	Low buffer / no listed financing	504	61.1 [56.3, 65.9]
2022	Other or unclassified	72	43.7 [30.2, 57.1]
2023	Cash/checking/savings	3,669	15.3 [14.0, 16.5]
2023	Bridge credit paid in full	4,116	8.7 [7.7, 9.6]
2023	Revolving credit for $400	1,687	47.2 [44.6, 49.9]
2023	Unable to pay $400	1,337	74.5 [71.9, 77.1]
2023	Low buffer / no listed financing	518	59.7 [54.9, 64.4]
2023	Other or unclassified	73	32.7 [20.7, 44.7]
2024	Cash/checking/savings	3,811	13.8 [12.6, 14.9]
2024	Bridge credit paid in full	4,630	9.2 [8.3, 10.1]
2024	Revolving credit for $400	1,724	47.4 [44.8, 49.9]
2024	Unable to pay $400	1,419	69.8 [67.2, 72.4]
2024	Low buffer / no listed financing	609	59.2 [54.8, 63.5]
2024	Other or unclassified	102	39.2 [28.9, 49.6]
2025	Cash/checking/savings	3,922	13.8 [12.7, 15.0]
2025	Bridge credit paid in full	4,846	8.3 [7.5, 9.1]
2025	Revolving credit for $400	1,833	47.6 [45.1, 50.1]
2025	Unable to pay $400	1,530	73.4 [71.0, 75.8]
2025	Low buffer / no listed financing	714	60.8 [56.9, 64.6]
2025	Other or unclassified	89	28.3 [18.3, 38.3]

Table 9 shows that the simplified profile ordering is not a one-year artifact. Bridge credit paid in full is consistently among the lowest-vulnerability groups; revolving credit is much higher; inability to pay is highest or near-highest; and low-buffer/no-listed-financing is elevated.

Table 10. Multi-year validation: simplified emergency-financing profiles and bill pressure, SHED 2023–2025. Bill pressure is the cleanest harmonized strict-hardship component available in the harmonized repeated cross-sections for 2023–2025.

Year	Profile	N	Bill pressure % [95% approx CI]
2023	Cash/checking/savings	3,669	8.2 [7.2, 9.2]
2023	Bridge credit paid in full	4,116	3.9 [3.3, 4.6]
2023	Revolving credit for $400	1,687	21.4 [19.3, 23.6]
2023	Unable to pay $400	1,337	48.8 [45.8, 51.8]
2023	Low buffer / no listed financing	518	39.8 [35.1, 44.5]
2023	Other or unclassified	73	25.6 [14.4, 36.7]
2024	Cash/checking/savings	3,811	7.8 [6.9, 8.7]
2024	Bridge credit paid in full	4,630	4.5 [3.9, 5.1]
2024	Revolving credit for $400	1,724	19.8 [17.8, 21.9]
2024	Unable to pay $400	1,419	46.0 [43.2, 48.9]
2024	Low buffer / no listed financing	609	41.4 [37.1, 45.8]
2024	Other or unclassified	102	25.2 [16.1, 34.4]
2025	Cash/checking/savings	3,922	14.6 [13.4, 15.8]
2025	Bridge credit paid in full	4,846	9.4 [8.5, 10.3]
2025	Revolving credit for $400	1,833	45.9 [43.5, 48.4]
2025	Unable to pay $400	1,530	72.9 [70.5, 75.3]
2025	Low buffer / no listed financing	714	65.6 [61.9, 69.3]
2025	Other or unclassified	89	43.8 [32.8, 54.9]

Table 10 uses bill pressure as the cleaner strict-hardship component available in recent harmonized years. The ordering again supports the main interpretation: bridge/cash profiles are lower-hardship than revolving, low-buffer/no-listed-financing, and inability-to-pay profiles. At the same time, the broad 2025 increase in bill pressure should be treated as a measurement-warning item until exact question wording and construction continuity are documented wave by wave.

11. Interpretation

The consolidated evidence supports four conclusions.

First, emergency-buffer indicators remain important. The post-2021 plateau/reversal in published SHED measures means liquid shock absorption did not keep improving after the temporary 2021 high-water mark.

Second, low-buffer adults are not homogeneous. Some use credit, some cannot use credit, and some rely on other borrowing or are unable to pay.

Third, credit use is too crude as a measurement category. Paid-in-full bridge credit is empirically distinct from revolving credit and other borrowing.

Fourth, the SHED lane is strongest when read as measurement infrastructure rather than causal identification. The contribution is a transparent classification of emergency-financing states that can support future policy analysis, dashboards, and individual decision tools.

12. Limitations

Several limitations should remain visible.

1. The analysis is descriptive. It does not estimate causal effects of savings, credit, or income on hardship.

2. Public-use inference is approximate because the inspected files do not expose public replicate weights, strata, or PSU variables.
3. The first-pass clustering validation is preliminary. Stronger versions could use latent class analysis, MCA, k-modes, Gower-distance clustering, or bootstrap stability checks.
4. The multi-year validation supports broad simplified ordering, not every fine 2025 profile category.
5. The pooled liquidity-exclusion/unclassified category is heterogeneous and should be interpreted cautiously.
6. The strict hardship composite combines bill pressure, outside help, and medical debt, which differ in timing and mechanism.
7. SHED reports adult responses; it is not a direct audit of household bank balances.

13. Review-response diagnostics

A first validation-upgrade pass adds four diagnostics: profile balance, component outcomes, classification sensitivity, and the 2025 bill-pressure jump. These diagnostics strengthen the measurement paper while preserving the central interpretation: the evidence remains descriptive and public-use inference remains approximate.

Appendix Table A1. Profile balance table, SHED 2025. Weighted adult shares within each emergency-financing profile. This table is descriptive and helps assess whether profile gradients may partly reflect composition rather than behavioral mechanisms.

Characteristic	Cash/checking/savings	Bridge credit paid in full	Revolving credit for $400	Other emergency borrowing	Unable to pay $400	Liquidity exclusion / unclassified
Income $100k+	46.2	61.1	28.5	14.3	5.9	12.0
Income <$50k	25.7	14.3	39.3	64.9	74.8	67.1
Bachelor's+	36.1	58.0	29.2	17.3	9.0	12.6
Age 60+	38.5	38.0	24.1	14.0	16.5	10.5
Renter	17.5	16.1	32.2	42.2	55.2	40.3
Working full-time	51.5	53.8	56.6	38.7	34.9	36.8
Has child under 18	25.0	24.8	34.1	40.4	39.0	43.7
White	65.8	71.0	49.1	44.4	40.6	35.6

Appendix Table A2. Component outcomes by profile, SHED 2025. Weighted rates with approximate final-weight/Kish confidence intervals. Component outcomes are shown separately so the strict composite does not hide different timing and mechanisms.

Profile	Strict hardship	Bill pressure	External help	Medical debt	Financially vulnerable	Credit constrained
Cash/checking/savings	31.4 [29.8, 33.0]	12.9 [11.7, 14.0]	14.4 [13.2, 15.6]	13.0 [11.9, 14.2]	12.0 [10.9, 13.1]	11.7 [10.6, 12.8]
Bridge credit paid in full	24.7 [23.4, 26.0]	8.7 [7.8, 9.6]	13.8 [12.8, 14.9]	8.1 [7.3, 8.9]	7.8 [7.0, 8.6]	6.2 [5.5, 7.0]
Revolving credit for $400	67.9 [65.6, 70.2]	45.9 [43.5, 48.4]	31.2 [28.9, 33.5]	32.4 [30.0, 34.7]	47.6 [45.1, 50.1]	30.8 [28.5, 33.0]
Other emergency borrowing	79.3 [76.5, 82.1]	56.0 [52.6, 59.4]	49.8 [46.4, 53.2]	28.2 [25.2, 31.3]	52.8 [49.4, 56.3]	36.1 [32.8, 39.4]
Unable to pay $400	85.1 [83.2, 87.0]	72.9 [70.5, 75.3]	42.0 [39.3, 44.7]	35.3 [32.8, 37.9]	73.4 [71.0, 75.8]	42.7 [40.1, 45.4]
Liquidity exclusion / unclassified	78.8 [72.2, 85.4]	62.8 [55.0, 70.7]	33.6 [25.9, 41.3]	27.1 [19.8, 34.3]	48.2 [40.1, 56.3]	32.3 [24.7, 39.9]

Appendix Table A3. Classification multiverse, SHED 2025. Strict-hardship gradients under alternative priority rules and non-exclusive response definitions. This is a first multiverse pass, not a final exhaustive coding-bounds exercise.

Specification	Group	N	Strict hardship % [95% approx CI]
Main priority / pooled	Cash/checking/savings	3754	31.4 [29.8, 33.0]
Main priority / pooled	Bridge credit paid in full	4704	24.7 [23.4, 26.0]
Main priority / pooled	Revolving credit for $400	1833	67.9 [65.6, 70.2]
Main priority / pooled	Other emergency borrowing	943	79.3 [76.5, 82.1]
Main priority / pooled	Unable to pay $400	1530	85.1 [83.2, 87.0]
Main priority / pooled	Liquidity exclusion / unclassified	170	78.8 [72.2, 85.4]
Bridge-first priority / pooled	Cash/checking/savings	4243	36.1 [34.5, 37.6]
Bridge-first priority / pooled	Bridge credit paid in full	5121	28.1 [26.8, 29.4]
Bridge-first priority / pooled	Revolving credit for $400	1236	71.2 [68.5, 73.9]
Bridge-first priority / pooled	Other emergency borrowing	645	80.9 [77.6, 84.1]
Bridge-first priority / pooled	Unable to pay $400	1530	85.1 [83.2, 87.0]
Bridge-first priority / pooled	Liquidity exclusion / unclassified	159	79.4 [72.6, 86.2]
Non-exclusive response	Cash/checking/savings response	5605	35.3 [34.0, 36.6]
Non-exclusive response	Paid-in-full card response	5137	28.3 [27.0, 29.6]
Non-exclusive response	Revolving card response	1936	69.4 [67.2, 71.6]
Non-exclusive response	Other borrowing response	1580	81.2 [79.1, 83.2]
Non-exclusive response	Unable-to-pay response	1530	85.1 [83.2, 87.0]
Mixed-response respondents	Selected two or more $400 financing responses	1895	43.8 [41.4, 46.2]

Appendix Table A4. Bill-pressure jump diagnostic, SHED 2023–2025. Weighted bill-pressure rates by simplified profile. The 2025 increase is broad-based and should be treated as a harmonization / measurement-warning item until exact questionnaire and coding continuity are fully documented.

Profile	2023	2024	2025	2025 minus 2024 pp
Cash/checking/savings	8.2 [7.2, 9.2]	7.8 [6.9, 8.7]	14.6 [13.4, 15.8]	6.8
Bridge credit paid in full	3.9 [3.3, 4.6]	4.5 [3.9, 5.1]	9.4 [8.5, 10.3]	4.9
Revolving credit for $400	21.4 [19.3, 23.6]	19.8 [17.8, 21.9]	45.9 [43.5, 48.4]	26.1
Unable to pay $400	48.8 [45.8, 51.8]	46.0 [43.2, 48.9]	72.9 [70.5, 75.3]	26.8
Low buffer / no listed financing	39.8 [35.1, 44.5]	41.4 [37.1, 45.8]	65.6 [61.9, 69.3]	24.2
Other or unclassified	25.6 [14.4, 36.7]	25.2 [16.1, 34.4]	43.8 [32.8, 54.9]	18.6

The balance table confirms that profile composition differs materially. Bridge-credit respondents are more advantaged on income and education than revolving-credit, other-borrowing, and unable-to-pay respondents. That does not weaken the taxonomy; it clarifies its interpretation. The taxonomy separates states that are compositionally and behaviorally different, but the resulting gradients should not be read as treatment effects.

The component-outcome table shows that the strict composite is not driven by a single item. Bill pressure, external help, medical debt, financial vulnerability, and credit constraint all show the same broad ordering, although they differ in timing and mechanism. Bill pressure is the cleanest contemporaneous outcome for future harmonized work.

The classification multiverse shows that the broad ordering survives the first alternative-coding pass. Paid-in-full bridge credit remains much closer to liquid sufficiency than to revolving credit, other borrowing, or inability to pay. Mixed-response respondents sit between liquid-sufficiency and distress profiles, which is exactly why a fuller coding-bounds exercise remains useful.

The bill-pressure diagnostic shows a broad 2025 increase across profiles. The repeated-cross-section ordering remains informative, but a journal-grade version should not lean on 2025 bill-pressure levels until an item-by-item harmonization appendix explains question wording, response options, and recoding continuity.

14. Conclusion

Emergency savings and $400 payment capacity are useful measures, but they are incomplete. A measurement system for adult respondents' household financial resilience also needs to know how adults expect to finance shocks. Paid-in-full bridge credit, revolving credit, other emergency borrowing, inability to pay, and liquidity exclusion are distinct states.

The consolidated SHED evidence supports an institutional measurement scaffold: distinguish liquid sufficiency, liquidity bridges, debt-financed distress, and liquidity exclusion; use strict non-overlap hardship outcomes when possible; validate profile ordering across coding choices and repeated cross-sections; and keep inference limits visible. This does not prove causality. It improves the map. For a research agenda focused on empowering individuals in personal economic decisions, that map is the point.

Selected references

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